



Muhammad Saood Sarwar

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EDUCATION AND TRAINING

2026/09/28 – CURRENT Nicosia, Cyprus

PHD IN COMPUTATIONAL SCIENCES The Cyprus Institute (Cyl)

- Doctoral research focuses on emergent phenomena in Generative AI (GenAI) models.
- Investigating the underlying mechanisms, behaviors, and capabilities that emerge in large-scale generative models.

2022/11/30 – 2025/11/30

MS, DATA SCIENCE National University of Computer and Emerging Sciences (NUCES) - FAST

Website <https://www.nu.edu.pk/> | **Field of study** Quantum Computing, Deep Learning, Quantum Machine Learning |

Final grade 3.97 / 4.0 |

Thesis A Quantum-Classical Hybrid Framework for Robust Medical Diagnostics via MCMC-Parameterised QUBO Optimization

2015/11/30 – 2019/11/30 Pakistan

BS, SOFTWARE ENGINEERING COMSATS University Islamabad, Attock Campus

Website <https://www.comsats.edu.pk> | **Field of study** Artificial Intelligence, Computer Vision | **Final grade** CGPA 3.8/4.0 |

Thesis Identification of rhythmic sounds patterns by Emotion Recognition using Landmark and Euclidean Distance techniques

2013/11/30 – 2015/11/30 Pakistan

HSSC, PRE- ENGINEERING APSACS Attock Cantt

Final grade A

2011/11/30 – 2013/11/30 Attock, Pakistan

SSC, SCIENCE Noble Grammar Public Secondary School

Final grade A+

WORK EXPERIENCE

MARIE SKŁODOWSKA-CURIE DOCTORAL RESEARCHER – FINALITY MSCA DOCTORAL NETWORK (THE CYPRUS INSTITUTE) – 2026/09/28 – Current – NICOSIA, CYPRUS

- Joined the FINALITY MSCA Doctoral Network as a Doctoral Researcher in Computational Sciences.
- Conduct doctoral research on emergent phenomena in Generative Artificial Intelligence (GenAI) model.
- Study the underlying mechanisms and characteristics of emergent behaviors in large-scale AI models

LECTURER (COMPUTING) – NATIONAL UNIVERSITY OF COMPUTER & EMERGING SCIENCES - FAST – 2026/07/31 – 2026/09/27 – ISLAMABAD, PAKISTAN

1. Lecturer (Computing) at FAST-NUCES Islamabad, responsible for teaching and academic instruction at the undergraduate level.
2. Engaged in research and academic activities in Quantum Computing, Quantum Machine Learning, and Artificial Intelligence.

Provided hands-on demonstrations and practical exercises to enhance students' understanding and problem-solving abilities.

INSTRUCTOR – NATIONAL UNIVERSITY OF COMPUTER & EMERGING SCIENCES - FAST PESHAWAR CAMPUS – 2023/07/31 – 2025/07/31 – PAKISTAN

Instructed students in fundamental computer science concepts, programming languages (e.g., Python, C++, C), algorithms, and data structures.

COMPUTER ENGINEER – GHULAM ISHAQ KHAN INSTITUTE OF ENGINEERING SCIENCES AND TECHNOLOGY – 2023/01/31 – 2023/06/30 – PAKISTAN

Holding help-lab hours for individual student consultations about assignments and lab tasks related to subject.

COMPUTER SCIENCE LECTURER – CADET COLLEGE SOHAWA PAKISTAN SWEET HOME – 2021/10/31 – 2023/01/31 – SOHAWA, PAKISTAN

Course instruction (lectures and labs) for programming languages in C and C++.

COMPUTER SCIENCE LECTURER – ARMY PUBLIC SCHOOL, DEFENSE COMPLEX ISLAMABAD – 2020/10/31 – 2021/09/30 – ISLAMABAD, PAKISTAN

Ability to provide the best practices to make the students experts in using computers.

SOFTWARE DEVELOPER – UNICO TECHNOLOGIES, ATTOCK – 2018/08/31 – 2020/06/30 – ATTOCK, PAKISTAN

- 1. Create basic website layouts and user interfaces with HTML and CSS.
- 2. Integrate data from various back-end web services and databases.

SKILLS

Python | Machine Learning | TensorFlow | Keras | Deep Learning | PyTorch | Quantum Computing | GenAI

LANGUAGE SKILLS

Mother tongue(s): URDU
Other language(s): ENGLISH

PROJECTS

Music Recommendation by Emotion Recognition using Landmark, Sobel-edges, and Convolutional Neural Network

- Developed a system for facial emotion recognition using landmark detection, Sobel-edges, and convolutional neural networks.
- Extracted landmark features from facial images to create a representation of the face that a CNN can process.

Datasets & Research Artifacts

AI vs. Real: 192-Class Scene Image Dataset

- Creator & Maintainer (with 1 collaborator)
- Curated, structured, and publicly released a large-scale dataset covering 192 scene categories with AI-generated and real-world images.
 - Designed as a benchmarking resource for AI-generated image detection, scene classification, and image authenticity verification.
 - Supports research in computer vision, deep learning, and synthetic media forensics.

- Published with documentation and metadata to enable reproducible and comparative research.

Link <https://www.kaggle.com/datasets/muhammadsaoodsarwar/ai-vs-real-192-class-scene-image-dataset>

● HONOURS AND AWARDS

2019/11/30

Gold Medalist at campus level – Comsats University Islamabad

2019/11/30

Silver Medalist at institute level – Comsats University Islamabad

2022/11/30

Business Analytics for Data-Driven Decision Making – Boston University

2020/11/30

Introduction to Internet of Things – Stanford University

● PUBLICATIONS

2026

Recurrent imputation network with patterned output (RINPO) for sparse data

Journal Name: Knowledge and Information Systems | **Volume, Issue and Pages:** Volume 68, article number 118 | **Publisher:** Springer Nature

Link <https://doi.org/10.1007/s10115-026-02708-2>

2025

Hybrid Quantum-Classical Machine Learning for Enhanced PCOS Detection.

Volume 7, article number 83, (2025)

Journal Name: Quantum Machine Intelligence | **Publisher:** Springer Nature

Link <https://doi.org/10.1007/s42484-025-00307-y>

2026

Deep Feature Extraction with Pretrained CNNs for Sorghum Leaf Disease Classification Using SVM

Journal Name: International Conference on Frontiers of Information Technology (FIT) | **Publisher:** IEEE

Link <https://doi.org/10.1109/FIT67061.2025.11333699>

2024

Enhancing Amazon Product Review Analysis by Using Bidirectional LSTM Neural Networks for Deep Multi-Domain Sentiment

Journal Name: 19th International Conference on Emerging Technologies (ICET) | **Publisher:** IEEE

Link <https://doi.org/10.1109/ICET63392.2024.10935260>

2023

Identification of Rhythmic Sounds Patterns by Emotion Recognition Using Landmark and Euclidean Distance Techniques

Journal Name: Journal of Image Processing and Intelligent Remote Sensing | **Volume, Issue and Pages:** Vol. 3 No. 03 (2023): April-May 2023

Link [10.55529/jipirs.33.27.35](https://doi.org/10.55529/jipirs.33.27.35)

● **RESEARCH INTEREST**

Research Interests in Generative AI, Quantum Machine Learning, Computer Vision, and Trustworthy AI.

- Quantum Machine Learning & Optimization
- Computer Vision & Synthetic Media Detection
- Deep Learning & Representation Learning
- Uncertainty-Aware and Robust Classification